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# Reliability of sitting posture between physical therapist video-based evaluation and SMART IMU system using rapid upper limb assessment (RULA)

Ratikanlaya Tanthuwapathom<sup>1,5</sup>, Udomporn Manupibul<sup>2,5</sup>, Wimonrat Jarumethitanont<sup>1,3</sup>, Weerawat Limroongreungrat<sup>4</sup>, Songpol Ongwattanakul<sup>1</sup> & Warakorn Charoensuk<sup>1</sup>✉

This study investigates the ergonomic assessment of sitting postures and the potential for work-related musculoskeletal disorders (WMSDs) in office environments by comparing traditional physical therapist evaluations with Inertial Measurement Unit (IMU) technology by determining the reliability and accuracy of sitting posture assessment using the rapid upper limb assessment (RULA) method. In this experiment, neck and body angle data is collected from twenty participants while sitting and working. The study aims to capture and compare the neck and trunk posture score based RULA protocol system to evaluate ergonomic risks. The findings revealed a strong correlation between the video-based evaluations by the physical therapist and the data obtained from the SMART IMU system, demonstrating the feasibility of using these combined approaches for ergonomic assessments. The results highlight the effectiveness of integrating traditional ergonomic assessment tools with modern IMU technology in comprehensively analyzing the ergonomic risks associated with prolonged sitting.

**Keywords** Work-related musculoskeletal disorders, Inertial measurement unit, Sitting posture, Rapid upper limb assessment

Currently, the increasing trend of using computers and smartphones affects health issues related to prolonged sitting in an office, particularly work-related musculoskeletal disorders (WMSDs). According to the world health organization (WHO), WMSDs refer to abnormal function of the musculoskeletal system associated with the work environment<sup>1</sup>. This includes issues arising from the body's organs involving movement, such as muscles, tendons, spinal bones, vertebral discs, blood vessels, bone tissue, and nerves. Musculoskeletal discomfort starts with a few atypical symptoms and might become persistent, causing impairment<sup>2</sup>. Due to repetitive muscular use, long periods of sitting (especially in an office or when using a computer) can cause pain. Sitting when working on a computer without changing posture may cause discomfort and fatigue in the arms, hands, and fingertips due to the peripheral nerves in each position being enforced<sup>3–5</sup>.

Prolonged sitting, especially in contexts such as office work or computer use, can lead to discomfort and pain due to sustained muscle contractions and poor posture. Two upper-body muscles significantly impacted during sitting are the upper trapezius and sternocleidomastoid. The upper trapezius, which extends from the neck to the shoulders and upper back, stabilizes the shoulder blades and supports the neck's weight. When the head is held forward during prolonged sitting, this muscle remains contracted to maintain head position, causing fatigue and discomfort. Similarly, the sternocleidomastoids, which run along both sides of the neck from the sternum and collar bone to the mastoid bone behind the ear, assist in neck rotation and bending. Poor posture while sitting, particularly with the head and shoulders hunched forward, can over activate these muscles, leading to tightness and neck pain. Piotr Kocur et al. found that forward head posture alone has little impact on muscle stiffness,

<sup>1</sup>Department of Biomedical Engineering, Faculty of Engineering, Mahidol University, Phuttamonthon, Nakhon Pathom 73170, Thailand. <sup>2</sup>Sirindhorn School of Prosthetics and Orthotics, Faculty of Medicine Siriraj Hospital, Mahidol University, Bangkok 10700, Thailand. <sup>3</sup>Faculty of Physical Therapy, Mahidol University, Phuttamonthon, Nakhon Pathom 73170, Thailand. <sup>4</sup>College of Sports Science and Technology, Mahidol University, Phuttamonthon, Nakhon Pathom 73170, Thailand. <sup>5</sup>Ratikanlaya Tanthuwapathom and Udomporn Manupibul contributed equally to this work. ✉email: warakorn.cha@mahidol.ac.th

tone, elasticity, or pressure sensitivity in healthy individuals. However, factors like cervical pain, musculoskeletal disorders, or prolonged sitting may play a role. Tightness in the upper trapezius and sternocleidomastoid muscles indicates sustained contraction and discomfort<sup>6</sup>.

Numerous studies have assessed the link between prolonged sitting and musculoskeletal pain. For example, a study published in the *Journal of Occupational Medicine and Toxicology*, Dang, T.H.A et al. explored the relationship between sedentary behaviors and musculoskeletal pain among German workers<sup>7</sup>. Using the Nordic Musculoskeletal Questionnaire, researchers identified a correlation between sitting duration and complaints of neck, shoulder, and back pain, highlighting the significant impact of prolonged sitting on workplace health. Another study published in the *Journal of Physical Therapy Science*, Putsa, B. et al. emphasized that sitting for more than 60 min without breaks is particularly detrimental, increasing discomfort and stiffness in the musculoskeletal system<sup>8</sup>. In general, sitting continuously for periods exceeding 30 to 60 min is considered detrimental to musculoskeletal health. Regular breaks, stretching, and adopting ergonomic postures are essential strategies to mitigate the risks associated with prolonged sitting.

Via the HSE's Historical Picture statistics in Great Britain in 2023, a survey of 1.8 million working people found that 26% had WMSDs<sup>9</sup>. Depending on working position, there are many methods for evaluating ergonomics. These evaluation methods are specifically designed to identify and understand musculoskeletal problems arising in the workplace. These methods include the rapid upper limb assessment (RULA), rapid entire body assessment (REBA), or other assessments based on work posture<sup>10–12</sup>.

There are many researchers focus on evaluating the poor sitting posture by using many assessment tools including RULA and wearable sensors. Lynn McAtamney and Nigel Corlett created RULA in 1993 to assess ergonomic posture by observing upper body movements. The assessment is divided into two main groups: First, group A evaluates the upper extremities: lower arm and wrist. Second, group B is neck, trunk, and leg assessments<sup>13–16</sup> with scores based on specified angles. This assessment can be used to evaluate ergonomics at work stations, such as workers' posture in factories or other workplaces. It is particularly helpful in determining the level of risk or danger associated with the activity that these workers do. Namwongsa S et al. examines the ergonomic risk assessment of a sample of people in their typical posture while using a smartphone using the RULA tool to assess the relationship between musculoskeletal disorders. It was found that the relationship between musculoskeletal disorders and ergonomic risks is statistically significant. The limitation is that experience-based evaluations were conducted only by licensed physical therapists<sup>17</sup>. Jang et al. study aimed to classify poor posture of the head and shoulders using wearable magnetic sensors and a deep learning algorithm. The craniovertebral angle (CVA) was used to analyze forward head posture and shoulder angle (FSA) was used to analyze rounded shoulder posture. From the experimental results, the accuracy for deep neural networks (DNN) and convolutional neural networks (CNN) was found to be 88.1 and 88.7%<sup>18</sup>. Jose Antonio Diego-Mas et al., using the Kinect sensor in observational methods for assessing postures at work, explore the potential of the Kinect range sensor in the ergonomic assessment of workplace postures. The development of a computerized Ovako Working Posture Analysis System (OWAS) was used to gather and process Kinect data. The results were compared to results obtained via human observer assessments. The study found that the result of classification is high accuracy. The limitations of the Kinect are its natural operating environment and that not being directly facing the sensor will reduce accuracy<sup>19</sup>. Yassary et al. conducted a study on the evaluation of head posture using an IMU. The study validated IMU with an inclinometer. A strong correlation was found ( $r=0.99$ ;  $p<0.001$ ) with cervical range of motion (CROM), which can be a reliable way to analyze head posture. The study also assessed head posture variation in healthy subjects and found that the IMU system accurately analyzed head posture. The device was cheap, easy to use, and could be used in various environments<sup>20</sup>. Xuzhong Yan et al. presented a wearable inertial measurement unit (WIMU) system for real-time motion warnings to prevent musculoskeletal disorders (WMSDs) in construction workers. It emphasizes the risk of lower back and neck pain due to unsafe postures in construction tasks. The system uses two IMUs to assess ergonomic risk in real-time and alerts the wearer via a connected smartphone app when the sensors detect dangerous postures. The study validates the system's effectiveness through lab and field experiments, demonstrating its potential to enhance self-awareness and self-management of WMSDs among workers<sup>21</sup>. Ahmed Humadi et al. examined the accuracy and repeatability of wearable IMUs to assess ergonomic risk through RULA in manual material handling tasks. This research compared 3D and 2D angle measurements against a motion-capture camera system. The results showed that the 2D method had lower error margins and was more consistent in estimating trunk and neck angles, thus being more reliable for RULA scoring. The limitations of this research are that it used a small sample size and only males. The experiment was conducted for a short period in each posture<sup>22</sup>.

Many researchers focus on the isolated operation method such as traditional ergonomic assessment tools like physical therapist evaluation-based RULA or innovative technologies such as IMU. A gap exists for research that integrates these tools to provide a more comprehensive assessment of ergonomic risks. Confirming the effectiveness of integrating traditional ergonomic assessment tools like video-based physical therapy with IMU for a more comprehensive ergonomic risk assessment involves conducting studies that apply both physical therapist evaluation and IMU technologies concurrently in settings and activities.

The contribution of this research is to investigate the reliability of the video-based physical therapist evaluation and SMART IMU system<sup>23</sup> using the RULA assessment protocol. The aim involves capturing both the detailed posture analysis and comparing subjective assessments of ergonomic risk factors associated with upper extremity MSD obtained by physical therapist with objective measurements of neck, shoulder, and trunk joint angles and upper body movements captured by SMART IMU during task performance by using intraclass correlation coefficient. The hypothesis is that the results of both methods would correlate in identifying ergonomic risks.

Materials and methods

The material part of this study is composed of participants, the SMART IMU device, RULA assessment, and webcam cameras. The method is composed of the experimental design and data analysis.

Participants

The study involves 20 healthy participants, consisting of 10 men and 10 women. Equation (1) was used to calculate the sample size, which indicated that a minimum of 10 participants would be necessary. However, the actual response rate exceeded expectations. As a result, 20 participants were included in the study, as a larger sample size is expected to enhance the statistical power of the analysis. These individuals are office professionals and Mahidol University students who use a desktop or laptop for at least six hours a day, four days a week. The participants must possess at least six months of computer experience. However, people with a history of neck, shoulder, or back injuries in the preceding six months, and those with medical conditions that affect the spine, muscles, bones, and joints in the upper back, such as spinal degeneration, scoliosis, and herniated discs, are excluded from the study. Table 1 displays the demographic data of twenty participants.

N = [(Zα + Zβ )/C]^2 + 3

α = 0.05, β = 0.2, C = 0.5 × ln (1 + r / 1 - r )

Expected correlation coefficient (r) = 0.8.

All participants are requested to sign a consent form according to the human ethics approval of Mahidol University (COA No. MU-CIRB 2021/148.2206). The informed consent was obtained to publish the informations and images in the online access publication.

Equipment

The equipment in this research consists of a device for measuring angles, a sitting posture assessment, and webcam cameras for recording the experiment.

SMART IMU

The SMART IMU system was developed by the SMART Laboratory (Biomedical Engineering, Mahidol University, Thailand)<sup>23</sup>. The IMU sensor comprises an accelerometer, gyroscope, and magnetometer integrated with a processing circuit to calculate angular data along the X, Y, and Z axes. Raw data from the SMART IMU sensors are filtered using a low-pass filter and a Kalman filter. The low-pass filter eliminates high-frequency noise, while the Kalman filter smooths the data to ensure accurate orientation measurements. The angles for the X and Y axes are derived from the filtered acceleration data, while the Z-axis angle is calculated using magnetic field data. Equations (2), (3), and (4) illustrate the calculations for the X, Y, and Z angles, respectively.

Angle X = arctan (-Gx / Gz)

Angle Y = arctan (Gy / sqrt(Gx^2 + Gz^2))

Angle Z = arctan (-Yh / Xh) + declination

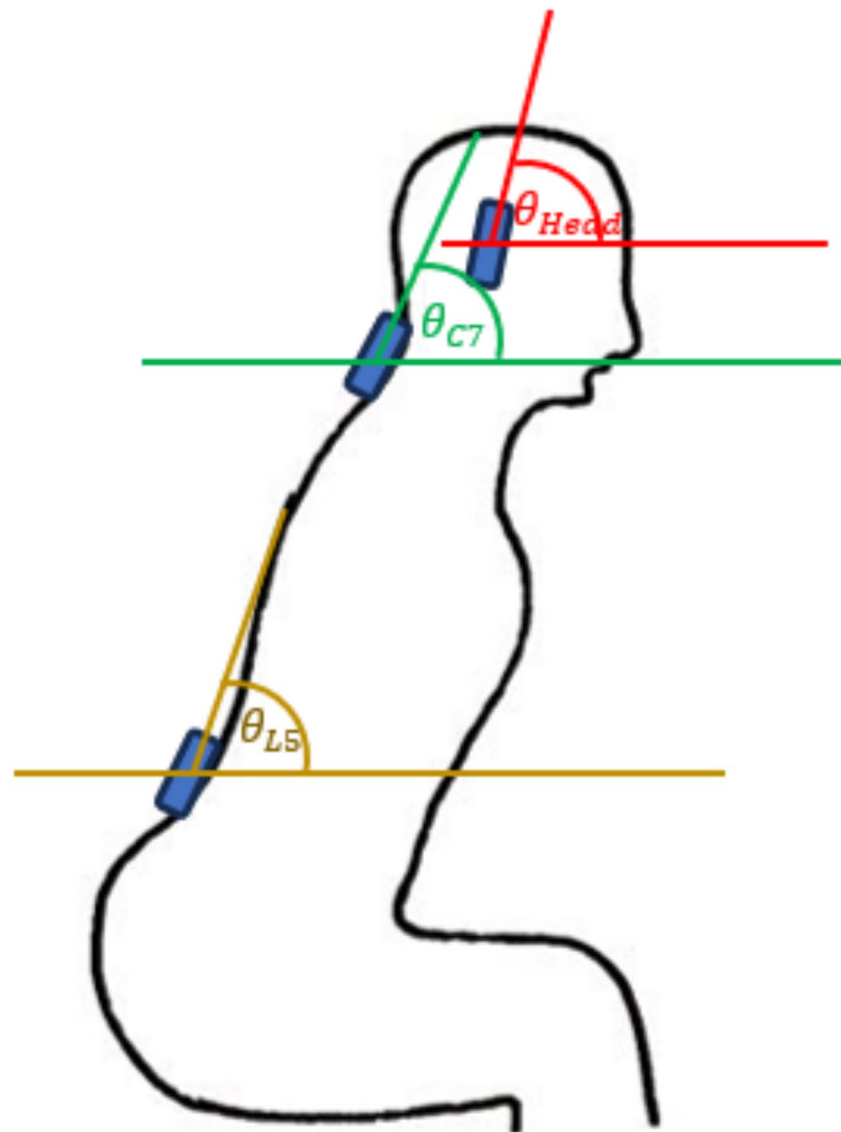
Where G<sub>x</sub>, G<sub>y</sub>, and G<sub>z</sub> are g units of acceleration data in X, Y, and Z axis. X<sub>h</sub> and Y<sub>h</sub> are the earth magnetics field in X and Y axis. The declination angle in any location can be found at <http://www.magnetic-declination.com/>.

In this study, three SMART IMU sensors were used to measure neck and trunk angles. The sensors were positioned on the upper body at the following locations: the right temporal area of the head, the seventh cervical vertebra (C7), and the fifth lumbar vertebra (L5), as shown in (Fig. 1). Each sensor recorded the absolute angular data for its respective segment. The neck angle was calculated as the relative angle between the head and C7, while the trunk angle was calculated as the relative angle between C7 and L5. Equations (5) and (6) outline these calculations. The system records the processed neck and trunk angle data at the sampling rate of 30 Hz.

θ Neck = θ C7 - θ Head

| Demographic dimension    | Range     | Mean   | Standard deviation |
|--------------------------|-----------|--------|--------------------|
| Age (years)              | 20–42     | 29.76  | 6.29               |
| Height (cm)              | 151–180   | 166.95 | 8.40               |
| Weight (kg)              | 42–99.6   | 68.56  | 16.04              |
| BMI (kg/m <sup>2</sup> ) | 16.6–31.7 | 24.40  | 4.44               |

Table 1. The demographic data of twenty participants.



$$\theta_{Trunk} = \theta_{L5} - \theta_{C7}$$

$$\theta_{Neck} = \theta_{C7} - \theta_{Head}$$

**Fig. 1.** Three sensor locations on upper body (head, C7, and L5).

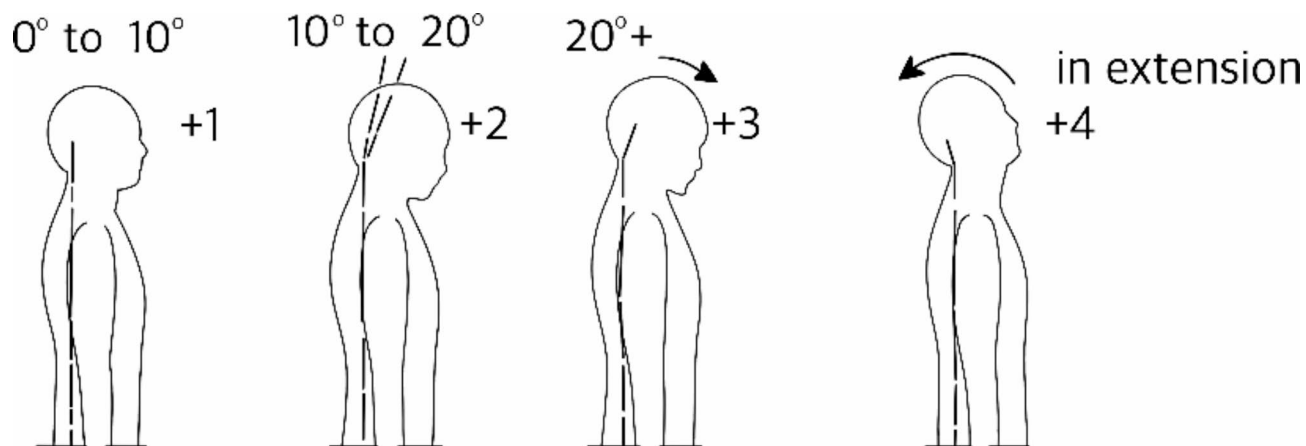
$$\theta_{Trunk} = \theta_{L5} - \theta_{C7} \quad (6)$$

#### *RULA assessment*

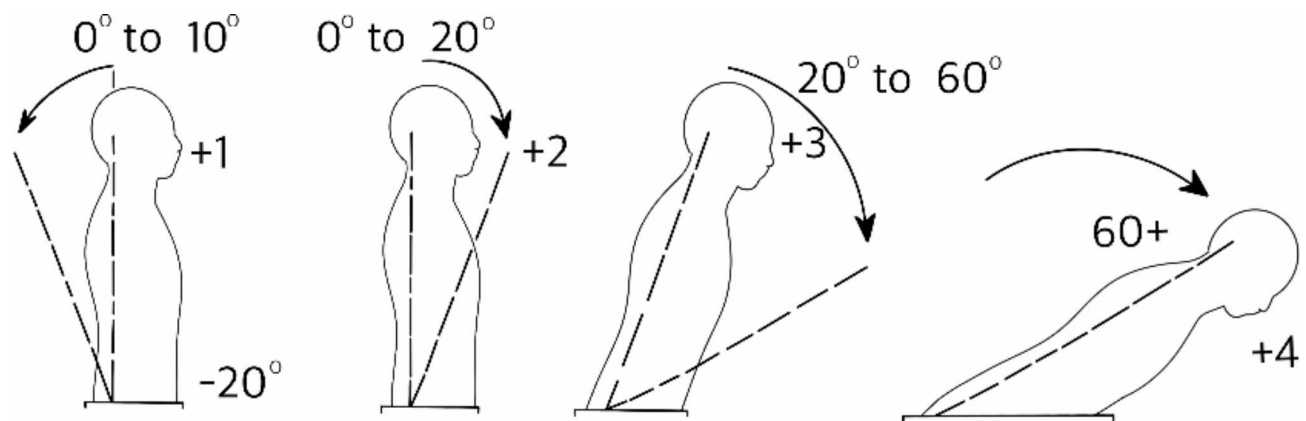
RULA is a survey method for evaluating the risk of musculoskeletal disorders (MSDs) and is used to observe the risk factors for upper limb abnormalities in workers. RULA uses a numerical scoring system to rate the body posture related to angle deviation. Neck and trunk parts of RULA are used to evaluate neck and trunk risk scores of sitting posture. The risk assessment from the score is used to interpret the risk level of WMSDs (Table 2). This

| Score | Level of WMSDs risk                             |
|-------|-------------------------------------------------|
| 1–2   | Negligible risk, no action required             |
| 3–4   | Low risk, change may be needed                  |
| 5–6   | Medium risk, further investigation, change soon |
| 6+    | Very high risk                                  |

**Table 2.** Interpretation of the level of MSD risk scores.



**Fig. 2.** The score for neck angle in the RULA assessment.



**Fig. 3.** The score for trunk angle in the RULA assessment.

experiment focuses on neck and trunk posture (Figs. 2 and 3). The scoring for neck and trunk posture is based on the criteria outlined in (Tables 3 and 4)<sup>13,24</sup>.

#### Webcams

Two webcams (Logitech C270 with 720p resolution and 30 frames per second) are set to record the movement of subjects on the frontal and sagittal sides.

#### Experimental design

The experiment are performed in the SMART laboratory (Mahidol University, Thailand). The neck and trunk joint angles are collected by the SMART IMU system with a sampling rate of 30 Hz. The video of the participants captured via the two webcams was evaluated by a physical therapist at a later stage.

Each participant is asked to score their neck, shoulder, and back pain from 0 to 10 before and after the experiment. During the experiment, a participant sits in the neutral position in front of the computer. Three IMU sensors are placed on the subject at the right temporal area of the head, the seventh cervical spine, and the fifth lumbar spine positions (Fig. 4). The SMART IMU system is calibrated and reset to eliminate any zero drift. In the neutral sitting position, each joint angle is set to zero degrees. Two webcams are set to record the

| Neck posture                     | Score |
|----------------------------------|-------|
| If neck flex in 0–10 degrees     | 1     |
| If neck flex in 11–20 degrees    | 2     |
| If neck flex more than 20 degree | 3     |
| If neck is an extension          | 4     |
| If neck is twisted               | +1    |
| If neck is side bending          | +1    |

**Table 3.** Neck posture scoring.

| Trunk posture                                                    | Score |
|------------------------------------------------------------------|-------|
| Trunk upright in<br>Also if trunk is well supported while seated | 1     |
| Trunk forward leaning in 0                                       | 2     |
| Trunk forward leaning in 21                                      | 3     |
| Trunk forward more than                                          | 4     |
| If trunk is twisted                                              | +1    |
| If trunk is side bending                                         | +1    |

**Table 4.** Trunk posture scoring.

sitting posture on the frontal and sagittal sides of subject. The SMART IMU system and webcam cameras record simultaneously. The environment is designed to be similar to work in an office.

The experiment is separated into two sessions:

1. Sitting in a neutral posture and typing a document on a desktop computer for 30 min and then taking a break for 10 min.
2. Sitting in a neutral posture and typing answers to a list of questions on the computer for 30 min.

The video recordings are used to evaluate the RULA score of neck and trunk parts by physical therapists. The neck and trunk angles from the SMART IMU system are converted to a RULA score following the RULA assessment. The scores based on RULA assessments from both the physical therapist and SMART IMU system are compared. To ensure that the ergonomic risk assessment accurately reflects real-world posture dynamics, the frequency and duration of different postures adopted during typing are considered. This comprehensive assessment allowed us to identify potential risks associated with the dynamic nature of typing, providing a more realistic evaluation of ergonomic risks.

**Data analysis**

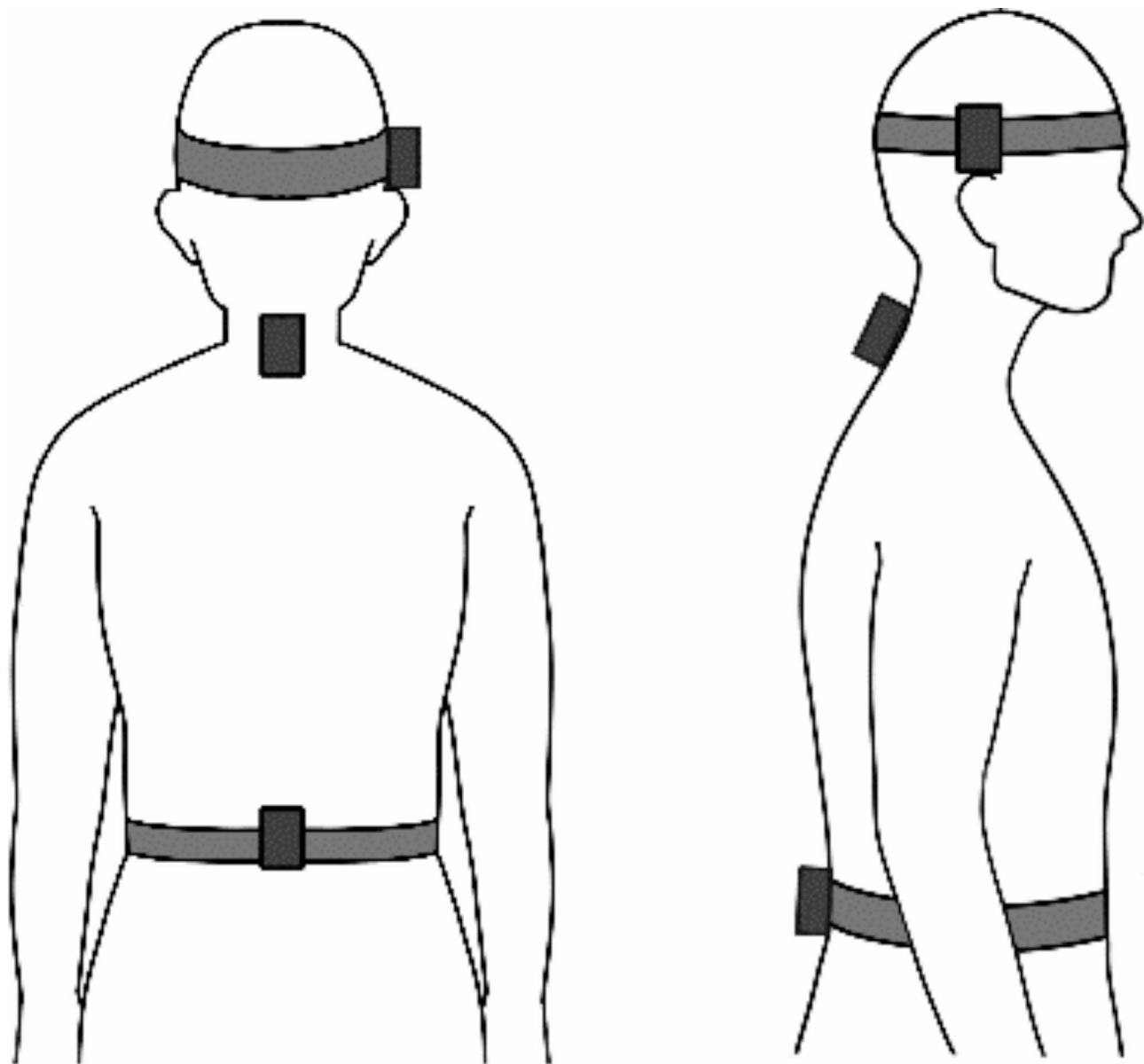
RULA assessment is the tool used for interpreting the data from video recordings and the SMART IMU system. This study utilized a 60 min data collection period to evaluate ergonomic postures. Before beginning the experiment, participants were instructed to sit still for one minute to establish a baseline measurement. Data analysis was conducted at 10-minute intervals, with samples taken at 0, 10, 20, 30, 40, 50, and 60 min. The focus was on key postures, including neck twisting, neck flexion, neck extension, neck side bending, trunk forward bending, trunk twisting, and trunk side bending, in accordance with the RULA assessment criteria. Movement data captured from video recordings, assessed by expert physical therapist, were compared with scores derived from IMU data to evaluate reliability and consistency between the two methods.

*Video recording analysis*

In the video recording interpretation, the physical therapist was asked to review the video recording and ascertain the score of the joint angle deviation from sitting posture following the RULA assessment. RULA scores of the neck angle and trunk angle are based on the flexion, extension, twisting, and side bending angles. Neck flexion is the motion of the chin toward the chest from a neutral position. Neck extension is the movement of the chin backward and away from the chest. Neck twisting is the small rotation of the neck that occurs between adjacent vertebrae. Neck side bending or lateral flexion is the movement of the neck to the lateral side. Trunk flexion is the movement of forward forward-bending of the spine which opposit direction to trunk extension. Trunk twisting movement is the rotation of the spine from the center of the body. Trunk side bending is the movement to the left and right of the trunk.

*SMART IMU system analysis*

The neck angle is obtained from SMART IMU devices at the temporal area of the head and the seventh cervical spine. The trunk angle is calculated from the seventh cervical spine and the fifth lumbar spine positions.



**Fig. 4.** SMART IMU sensors attachment positions.

#### *Evaluation of the neck score*

If the head is straight or slightly bent forward (the head's angle is not more than 10 degrees from the vertical or the neck's axis), the score will be 1 point. If the head is flexed forward to the vertical between 10 and 20 degrees, the score will be 2 points. If the head is flexed forward greater than 20 degrees to the vertical, the score will be 3 points. If the head is extended, the score will be 4 points. If the neck is twisting or bending, the score is increased 1 point.

#### *Evaluation of the trunk score*

If the trunk extends 0–10 degrees, the score will be 1 point. If the trunk flexes 0–20 degrees, the score will be 2 points. If the trunk moves forward between 20 and 60 degrees, the score will be 3 points. If the trunk moves forward greater than 60 degrees, the score will be 4 points. If the trunk is twisting or bending, the score is added one point.

| Reliability                 | ICC  | 95% CI    |
|-----------------------------|------|-----------|
| Locate neck position        | 0.89 | 0.87–0.91 |
| Neck twisted position       | 0.91 | 0.89–0.93 |
| Neck side bending position  | 0.92 | 0.90–0.94 |
| Neck Score                  | 0.91 | 0.89–0.93 |
| Locate trunk position       | 0.84 | 0.85–0.90 |
| Trunk twisted position      | 0.89 | 0.87–0.91 |
| Trunk side bending position | 0.84 | 0.85–0.90 |
| Trunk score                 | 0.84 | 0.85–0.90 |

**Table 5.** The intra-class correlation coefficients of neck and trunk angle score. ICC is intra-class correlation coefficient.

Results

The reliability of the equipment was analyzed by comparing the score results from the physical therapist and the SMART IMU system. SPSS software is used to calculate the correlation coefficient. Table 5 shows the intra-class correlation coefficients of neck and trunk angle scores. The results from the SMART IMU system and physical therapist show strong correlations.

Discussion

The primary objective of this study is to compare the reliability of ergonomic assessments conducted using the SMART IMU system against those performed by video-based physical therapist using the RULA method. The results from Table 5 indicate that IMU technology demonstrates a high degree of reliability, close to the consistency achieved through traditional physical therapist assessments. The high reliability of SMART IMU-based assessments, as evidenced by the Intraclass Correlation Coefficient (ICC). The neck score compared between the two systems had an ICC 0.89–0.92 which is interpreted as a very strong correlation<sup>25–31</sup>. The trunk score evaluation had an ICC 0.84–0.89, also indicating a strong correlation. Our findings highlight not only the potential for enhanced ergonomic assessment but also the challenges and considerations that accompany the integration of these tools.

The WMSD risk assessment study indicated that RULA is the most popular sitting posture examination<sup>11,15,32–37</sup>. The assessment model contains specific angles for scoring to analyze the risk of work postures generating WMSDs. It can evaluate the appropriateness of posture while working which makes it applicable to many types of work. Video-based RULA’s weakness is that it still requires ergonomic experts. The evaluation score depends on their expertise, which may still be unreliable. The study of Namwongsa S et al.<sup>17</sup> also mentions the need for a physical therapist’s experience to use RULA for ergonomic evaluation.

The novelty of this study lies in its integration of traditional ergonomic assessment tools (video-based physical therapist evaluation) with modern IMU technology. While both approaches have individual benefits, their combination provides a more comprehensive and objective assessment of ergonomic risks, addressing the limitations of each method when used in isolation, particularly the reliance on physical therapists’ expertise in using RULA. This integration allows for a more detailed analysis of sitting posture and upper body movements, potentially leading to more effective interventions for preventing WMSDs.

The SMART IMU system was developed for motion analysis, and it was applied to evaluate sitting positions when working on a computer in this research. A factor that affects the accuracy of SMART IMU devices in measuring angles is the error that may occur during the calibration process. While doing the calibration, the subject is requested to stay still in the neutral sitting position. The error may occur if the subject is moving. Some researchers such as Wonil Lee et al. also found limitations to using IMU sensors which were related to the size and placement of the sensors. They investigate the effectiveness of wearable sensors for analyzing trunk posture in construction workers. It compares single-parameter monitoring wearable sensors (SPMWS) and multi-parameter monitoring wearable sensors (MPMWS) placed at different body locations during simulated construction tasks. The study found that sensor placement significantly affects the accuracy of trunk posture analysis, with chest and back placements providing the most reliable data. The research suggests that wearable sensors can offer valuable insights into ergonomic risk in construction, but the choice of sensor type and placement is crucial for accurate posture analysis. The limitation of this study is the controlled laboratory setting, which may only partially replicate natural environment construction activities<sup>38</sup>.

This study highlights the potential of the SMART IMU system for ergonomic assessment, demonstrating its high reliability compared to traditional video-based RULA evaluations. However, several limitations should be considered. First, the study was conducted in a controlled laboratory environment, using only a single desk type. While this setup ensured consistent measurements, it does not reflect the variability of real-world office settings, where factors such as inconsistent desk height, inappropriate chair dimensions, and individual physical characteristics can significantly influence sitting posture and muscle group usage. Additionally, the experiment’s observation period was limited to just one hour, which may be insufficient to capture long-term changes in posture or the onset of fatigue, especially in healthy individuals who may adopt different postures over extended periods.

Another limitation is the controlled nature of the study, which excluded many dynamic and uncontrollable factors present in real-world office settings, such as environmental distractions or spontaneous movements.

Moreover, the analysis focused solely on angular measurements and posture evaluation without incorporating complementary tools such as electromyography (EMG) or 3D motion capture. This restricts the ability to assess muscle fatigue, tension, and pain caused by poor sitting postures, such as rounded shoulders or forward head positions, which could enhance the understanding of ergonomic risks.

Pose estimation algorithms like OpenPose and MediaPipe present significant potential for estimating body angles from video recordings, offering several advantages. These methods are non-invasive, as they do not require attaching sensors to the body, which enhances comfort and allows for natural movement during assessment<sup>39</sup>. Additionally, they are often more cost-effective and provide visual representations of body posture, aiding in the understanding of movement patterns and the identification of ergonomic risks. Incorporating a comparison between pose estimation and IMU-based methods within future studies could enhance the relevance of the research, providing valuable insights into the strengths and limitations of each approach. Factors such as accuracy, cost, invasiveness, and data visualization capabilities should be considered, contributing to a comprehensive understanding of ergonomic assessment tools and their optimal application across different settings.

Building on this, future research should address current limitations and explore the integration of IMU technology with other advanced analysis tools. Studies conducted in real-world office environments with diverse furniture setups and extended observation periods would provide more practical insights. Combining IMU-based assessments with electromyography (EMG) could further enhance the evaluation of muscle fatigue and identify postures that lead to discomfort or pain. Additionally, integrating 3D motion capture technology could offer detailed analyses of body movements and their ergonomic implications. These advancements, alongside comparisons with pose estimation methods, could guide the redesign of work environments and industrial facilities, ultimately improving quality of life and reducing the prevalence of musculoskeletal disorders (MSDs) through more effective risk identification and intervention strategies.

## Conclusion

This research contributes to the validation of the subjective method (video-based physical therapist) with the objective method (SMART IMU system) based on the RULA assessment protocol. The results show there are no significant differences between evaluations from physical therapist and the SMART IMU system. The integration of both the subjective method and objective method can enhance the effectiveness of the evaluation which may contribute to the prevention of work-related musculoskeletal disorders. In future studies, IMU system-based RULA assessment can be applied with other analysis tools such as electromyography (EMG) and 3D motion capture to advance the practice of ergonomic risk assessment. The research can be applied to redesign working environments and industries including office settings and working facilities for better quality of life. The research lays the groundwork for future innovations in ergonomic assessment, with the ultimate goal of reducing the prevalence of MSDs in the workplace through more effective risk identification and intervention strategies.

## Data availability

The data that support this study are available from the corresponding author upon reasonable request.

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## Author contributions

RT and UM are equal contributors to this work and are designated as co-first authors. RT, UM, WJ, WL, SO and WC contributed to the concept of working and research design. RT and UM designed the experiments. RT, UM, and WJ performed the experiments and analyzed the data. RT, UM, WC and WJ wrote the main manuscript texts including introduction, material and method, results, discussion, and conclusion. RT and UM prepared tables and figures. All authors reviewed the manuscript.

## Declarations

## Competing interests

The authors declare no competing interests.

## Ethics approval and consent to participate

The study protocol was approved by the Mahidol University Central Institutional Review Board (reference number: COA No. MU-CIRB 2021/148.2206), and the study was conducted in accordance with the Declaration of Helsinki.

## Consent for publication

The informed consent was obtained from all subjects and/or their legal guardian(s).

## Additional information

**Correspondence** and requests for materials should be addressed to W.C.

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